Ch 13: Graphical User Interface (GUI) Programming and tkinter Module

Application Program Development

Derek Harter

Department of Computer Science
East Texas A&M University

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Graphical User Interfaces (1 of 3)

Graphical User Interface (GUI)

A graphical user interface allows the user to interact with the operating system and other programs using graphical elements such as icons, buttons and dialog boxes.

- **User interface**: the part of the computer with which the user interacts
- **Command line interface**: displays a prompt and the user types a command that is then executed
- **Graphical User Interface (GUI)**: allows users to interact with a program through graphical elements on the screen

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Graphical User Interfaces (2 of 3)

- Command line interfaces

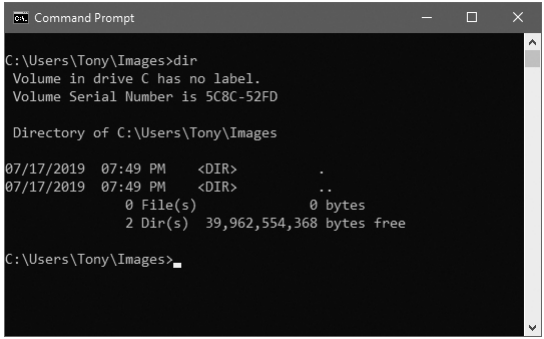


Figure 1: A command line interface (Gaddis, 2021, pg.652)

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Graphical User Interfaces (3 of 3)

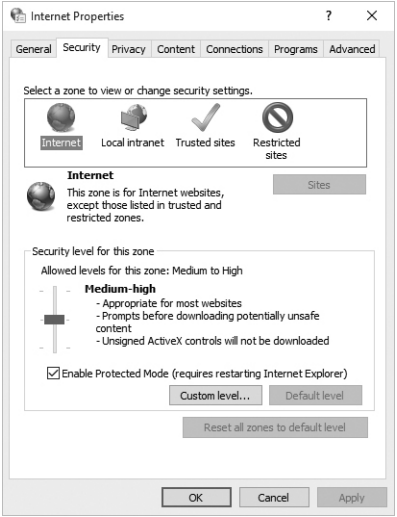


Figure 2: A dialog box (Gaddis, 2021, pg.652)

- **Dialog boxes**: small windows that display information and allow the user to perform actions
  - Responsible for most of the interaction through a GUI
  - User interacts with graphical elements such as icons, buttons and sliders

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GUI Programs are Event-Driven

- In text-based (command line driven) environments, programs determine the order in which things happen
  - The user can only enter data in the order requested by the program (e.g. sequence of prompts)
- GUI environments is **event-driven**
  - The user determines the order in which things happen
  - User causes events to take place and the program responds to the events

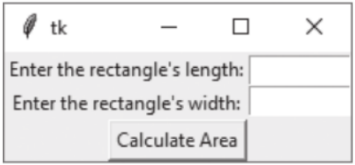


Figure 3: A basic Python tkinter GUI program (Gaddis, 2021, pg.653)

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Using the Python `tkinter` Module (1 of 3)

- No GUI programming features built into Python
- **`tkinter` module**: allows you to create a simple GUI program
  - Comes with Python
- **Widget**: a graphical element that the user can interact with or view
  - Presented by a GUI program

Using the Python `tkinter` Module (2 of 3)

Table 1: Some `tkinter` widgets

Widget	Description
`Button`	A button that can cause an action to occur when it is clicked.
`Canvas`	A rectangular area that can be used to display graphics.
`Checkbutton`	A button that may be in either the "on" or "off" position.
`Entry`	An area in which the user may type a single line of input from the keyboard.
`Frame`	A container that can hold other widgets.
`Label`	An area that displays one line of text or an image.
`Listbox`	A list from which the user may select an item.
`Menu`	A list of menu choices that are displayed when the user clicks a `Menubutton` widget.
`Menubutton`	A menu that is displayed on the screen and may be clicked by the user.
`Message`	Displays multiple lines of text.
`Radiobutton`	A widget that can be either selected or deselected. `Radiobutton` widgets usually appear in groups and allow the user to select one of several options.
`Scale`	A widget that allows the user to select a value by moving a slider along a track.
`Scrollbar`	Can be used with some other types of widgets to provide scrolling ability.
`Text`	A widget that allows the user to enter multiple lines of text input.
`Toplevel`	A container, like a `Frame`, but displayed in its own window.



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Using the Python `tkinter` Module (3 of 3)

- Programs that use `tkinter` do not always run reliably under IDLE (or in containers).
  - For best results run them from the operating system command prompt
  - For containers, may need to ensure that external display of gui widgets (X windows config) is configured properly
- Most programmers take an object-oriented approach when writing GUI programs
  - `__init__` method builds the GUI
  - When an instance is created the GUI appears on the screen

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Display Text with `Label` Widgets

- **`Label` widget**: displays a single line of text in a window
  - Made by creating an instance of `tkinter` module's `Label` class
  - Format: `tkinter.Label(self.main_window, text='my text')`
  - First argument references the root widget, second argument shows text that should appear in label
- **`pack()` method**: determines where a widget should be positioned and makes it visible when the main window is displayed
  - Called for each widget in a window
  - Receives an argument to specify positioning
    - Positioning depends n the order in which widgets were added to the main window
    - Valid arguments: `side='top'`, `side='left'`, `side='right'`



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Adding Borders to Labels

- When creating a `Label` widget, you can use the `borderwidth` and `relief` arguments to display a border around the label
- The `borderwidth` argument specifies the width of the border in pixels
- The `relief` argument specifies the border style



Figure 4: Examples of different styles of borders (Gaddis, 2021, pg.661)

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Internal and External Padding (1 of 2)

- **Padding**: space that appears around a widget
  - **Internal padding** appears around the inside edge of a widget
  - **External padding** appears around the outside edge of a widget

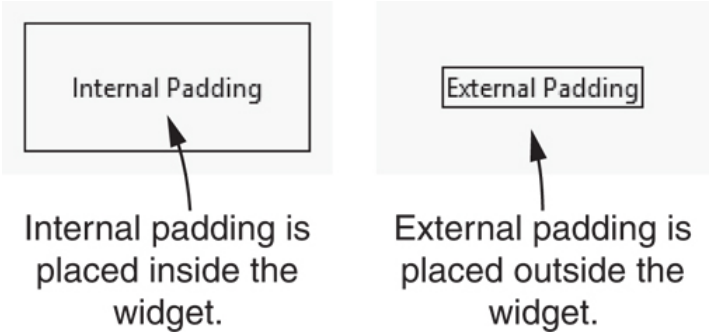


Figure 5: Internal and external padding (Gaddis, 2021, pg.662)

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Internal and External Padding (2 of 2)

- To add horizontal internal padding to a widget, pass the argument `ipadx=n` to the widget's pack method
- To add vertical internal padding pass the argument `ipady=n` to the widget's pack method

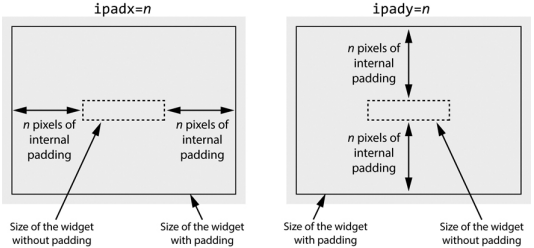


Figure 6: Horizontal and vertical internal padding (Gaddis, 2021, pg.662)

- To add horizontal external padding to a widget, pass the argument `padx=n` to the widget's pack method
- To add vertical external padding pass the argument `pady=n` to the widget's pack method

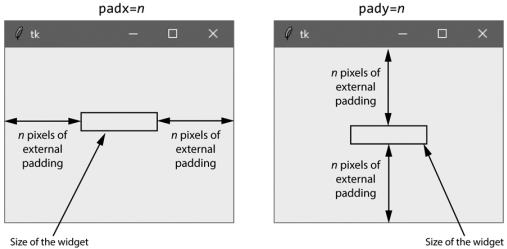


Figure 7: Horizontal and vertical external padding (Gaddis, 2021, pg.664)

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Summary

- The **user interface** is the part of the computer with which the user interacts
- A **graphical user interface** or **GUI** allows users to interact with a program through graphical elements on the screen
- GUIs are inherently **event-driven**, this means that the user determines the order and timing of when to interact with and cause things to happen
  - This requires an **event-loop** with callbacks to handle GUI interface interactions
- The `tkinter` module provides basic GUI elements and is available in the base Python packaging system
  - There is a more modern set of themed user interface components that is part of the base tkinter See tkinter.ttk (Org, 2026)
  - But there are more modern looking extensions, see customtkinter (Schimansky, 2026)
- Most programmers prefer to take an object-oriented approach when writing a GUI program
  - For Python `tkinter` this usually means creating GUI elements in `__init__` method
  - as well as usually starting the main event loop after GUI elements are created
- Use the `tkinter` `Label` widget to display text in a window
  - You arrange widgets by calling `pack()` with various positioning arguments
  - Order you add widgets to a container will effect layout (sometimes, e.g. see grids)



References

- Gaddis, T. (2021). *Starting out with Python* (5th). Pearson.
- Org, P. (2026). *tkinter Python interface to Tcl/Tk*. Retrieved November 2, 2026, from https://docs.python.org/3/library/tkinter.html
- Schimansky, T. (2026). *CustomTkinter: A modern and customizable Python UI-library based on Tkinter*. Retrieved October 30, 2026, from https://customtkinter.tomschimansky.com/

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